

SEQUENCES, SERIES AND DIFFERENTIAL EQUATIONS

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What we have learned in the previous lecture

- Introduction to differential equations
 - Ordinary differential equations (ODE): derivatives with respect to one variable only

$$a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \dots \\ + a_2(x)y''(x) + a_1(x)y'(x) + a_0(x)y(x) = f(x).$$

- If $f(x)=0$, the ODE is said to be homogeneous, otherwise non-homogeneous.
- Solving first order differential equations
 - Separable equations

$$\frac{dy}{dx} = f(x)g(y),$$



$$\frac{dy}{g(y)} = f(x) dx$$



Today's agenda

- First order differential equations
 - Non-homogeneous equations
- Second order differential equations
 - Equations reducible to first order
 - General form

Solving first order differential equations

- First order linear equations

$$\frac{dy}{dx} + p(x)y = q(x),$$

The equation is omogeneous in case $q(x)=0$. Then we have:

$$\frac{dy}{y} = -p(x) dx,$$

Which is separable and can be solved by integrating both sides.

Solving first order differential equations

- There are two methods for solving the nonhomogeneous equation

$$\frac{dy}{dx} + p(x)y = q(x),$$

- Method 1: using integrating factors

$$y(x) = e^{-\mu(x)} \int e^{\mu(x)} q(x) dx.$$

where $\mu(x) = \int p(x) dx$

- Method 2: variation of the parameter. Starting from the solution of the corresponding homogeneous equation $y = K e^{-\mu(x)}$, and replacing the constant by an unknown variable $k(x)$ $\longrightarrow k'(x) = e^{\mu(x)} q(x)$.
Integrating the right side brings to the solution of $k(x)$, and therefore y .

Differential equations of the second order

- Second order ordinary differential equations take the form

$$F\left(\frac{d^2y}{dx^2}, \frac{dy}{dx}, y, x\right) = 0$$

- Equations reducible to first order

$$F\left(\frac{d^2y}{dx^2}, \frac{dy}{dx}, x\right) = 0$$

By setting $v=dy/dx$, we obtain

$$F\left(\frac{dv}{dx}, v, x\right) = 0. \quad \longrightarrow \quad y = \int v(x) dx$$

$$F\left(\frac{d^2y}{dx^2}, \frac{dy}{dx}, y\right) = 0$$

By setting $v=dy/dx$, $\frac{d^2y}{dx^2} = \frac{dv}{dx} = \frac{dv}{dy} \frac{dy}{dx} = v \frac{dv}{dy} \longrightarrow F\left(v \frac{dv}{dy}, v, y\right) = 0$, first order equation of v as a function of y

Differential equations of the second order

- The general form of second-order **linear** differential equation is

$$a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = f(x).$$

- Corresponding homogeneous equation:

$$a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$$

It has a general solution: $y_h = C_1y_1(x) + C_2y_2(x)$,

Whenever a solution is known, another independent solution (and therefore the general solution) can be found by substituting $y = v(x)y_1(x)$ in the general equation.